

Is Clean Data All You Need?

A Search for the Best Domain Adaptation

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WS 2025-26

This is a term project report for the course *Is Attention All You Need? The Search for a New Architecture*, taught by Dr Michael Staniek. Our code implementation is available on GitHub.¹

1 Introduction

For this project, we adapt general-domain Transformers with non-standard attention to the medical domain. Within training, we explore how bidirectional and causal models react to different versions of the same dataset:

- Original data that contains some ill formatting.
- Refined data of the same content but well-formatted.

Afterwards, we evaluate the adapted models with standard metrics and compare standard attention models with those using non-standard attention. Therefore, we have two baselines (models with standard quadratic attention; original dataset) and two treatments (models with non-standard attention; cleaned dataset).

We suppose the following hypotheses for our experiments:

- Given the limited size of our dataset and its contexts, the models with standard and non-standard attention show neither significant difference in runtime nor perplexity.
- Cleaned dataset yields the lowest perplexity across all models.

2 Data

We use a medical-domain dataset because it is outside of the general domain, so we can rest assured that the models were not trained on it beforehand.

2.1 MedMCQA

The MedMCQA dataset Pal et al. (2022) consists of human-authored multiple-choice questions from entrance exams to All India Institute of Medical Sciences (AIIMS) colleges and the National Eligibility cum Entrance Test for Postgraduates (NEET PG). It contains 194 thousand data entries.

Each entry has a question, 4 answer options, the index of the correct option, an explanation that justifies the correct answer, and the subject to which the question belongs. We find all content attributes to be unreliable in formatting. To be precise, they have irregular spacing and noticeable PDF OCR artefacts leading to spelling mistakes. Furthermore, questions might lack question words and question marks, appearing more like keywords than sentences. Some entries have no explanations (50% in dev, 12% in train, and 100% in test). The existing explanations often reveal the correct answer option, are very short (under 100 characters), and provide too little relevant information to reach the correct option when

¹<https://github.com/bohdana-ivakhnenko/knowledge-conflicts>

answering the question. We have also noticed duplicate entries and duplicate answer options. See a full example in Table 1.

Attribute	Description	Example
question	Original question.	Hypomimia is?
question_upd	Cleaned question with proper formatting.	What is hypomimia?
exp	Explanation of the correct answer. In many entries, the explanation is absent or very short, leaving us with around 10% valid explanations.	Ans. C. Deficit of expression by gesture. Hypomimia or amimia is a deficit or absence of expression by gesture or mimicry. This is usually most obvious as a lack of facial expressiveness mobility (mask-like facies). This is a feature of frontal subcortical disease.
exp_upd	Cleaned explanation with fixed formatting issues and framed as stand-alone context, with meaning intact.	Hypomimia, also known as amimia, is a deficit or absence of expression by gesture or mimicry. This is typically most apparent as a lack of facial expressive mobility, often resulting in a “mask-like” appearance.
opa opb opc opd	Original answer options	• Decreased ability copy • decreased execution • Deficit of expression by gesture Deficit absence • Deficit of fluent speech
opa_upd opb_upd opc_upd opd_upd	Cleaned answer options with fixed formatting and spelling mistakes, while the terms stay the same, even if outdated.	• Decreased ability to copy • Decreased execution • Deficit of expression by gesture • Deficit of fluent speech
cop	Index of the correct answer option	3 (i.e., C)
subject_name	The medical subject to which the entry belongs.	Psychiatry

Table 1: Attributes of the resulting MedMCQA dataset with explanations and examples.

2.2 Cleaning Pipeline

We address some of the issues mentioned in the dataset description by cleaning questions, answers, and explanations using the `meta-llama/Meta-Llama-3-8B-Instruct` model, thus creating a cleaned version of the dataset. For this project, only explanations are directly relevant, but since we also use this dataset in the CrossTemporal NLP project (Ivakhnenko et al., 2026), we clean it fully.

Each cleaning task is introduced to the model with a specialised system prompt. All system prompts include an editor persona (Xu et al., 2025), inform the model what it will receive, list a clear set of cleaning instructions, and provide several positive and negative demonstrations (Khalifa et al., 2023). All examples were taken from the data (the specific IDs are mentioned in the repository, see `utils/prompts/examples.py`). Before cleaning, all data entries were numbered for easier navigation in the following order: train, dev, test.

2.2.1 Questions

The questions were cleaned without filtering to follow a standard structure. For cleaning, we used the prompt displayed in Figure 1, accompanied by five examples in Table 2.

Since many of them were so concise that they lacked the necessary information to derive the appropriate question word, the model was also provided with answer options. This introduced the problem of potential prompt-leaking. Additional complications arise because some questions include context that the model is tempted to cut out. Furthermore, the dataset features two types of questions: single- and multi-choice. The latter includes a list of options within the question, and the answer options consist of the combination of references to the options in the list (e.g., list of options: "[Question] a) option 1, b) option 2, ... e) option 5"; option list: "A. ab, B. bcd, C. be, D. abd"). Such an approach proved confusing both to humans working on the project and the model cleaning the data, so we also used a prompt specialised for this question type (see `utils/prompts/cleaning/question_op_list_disappeared_ids.txt`).

We use the following heuristics for automatic detection of problems in cleaned questions:

- The original question and the model output are different in length by two times or more.
- The model output includes comments from the model.
- The model output contains answer options, while the original has none.
- The original output contains answer options, but they disappear from the model output.

In these cases, the model is set to regenerate up to 10 times. The comments are additionally removed afterwards.

2.2.2 Answers

Even though hard to spot without domain knowledge, some answers also have issues. For example, *Vin* and *Yang* (instead of *Yin* and *Yang*). The prompt we use is shown in Figure 2 and uses five examples from Table 3. The model is provided with additional context through original (non-cleaned) questions to ensure higher quality.

For this task, we had the simplest automatic quality control. Each model output should include the required options: A, B, C, D. Otherwise, the model is asked to generate again up to 10 times.

2.2.3 Explanations

Not all data entries include explanations (17218 cases are missing in `train`), but for those that do, the explanations can be lengthy, increasing the time and memory required to process them. To optimise resource use, they were first classified into well-formatted (`exp_to_edit=False` for 53540 cases in `train`) and ill-formatted (`exp_to_edit=True` for 65175 cases in `train`) with the prompt in Figure 3 and three demonstrations. Then, only those explanations that require editing are cleaned. The cleaning task is performed using the prompt in Figure 4. It is accompanied by four demonstrations in Table 4. Only the explanations themselves are provided as input to the model, as they have sufficient context for editing.

The quality of cleaning is assessed automatically with the following heuristics:

- The original explanation and the model output are different in length by two times or more.
- The model output includes comments from the model.

In such cases, the output is regenerated up to 10 times. If the comments are still present, they are removed with rules (however, after running our experiments, we have discovered that the final version of the training data still includes comments; we suspect they slipped through in the chaos of data joining, which can be observed in `data/MedMCQA_processing.ipynb`).

All questions and answers were cleaned, while explanations were first classified as needing cleaning to save time, since they can be longer.

A prompt with a persona was developed and tweaked for each use case (see Figure 1). Perplexity was used to streamline the logic and ensure that the instructions are not inconsistent. Additionally, task-specific demonstrations were added.

INPUT		OUTPUT
Original Question	Answer Options	Cleaned Question
Characteristic of venous blood flow of lower limb in duplex Doppler is?	Monophasic Biphasic Triphasic Non phasic	What is characteristic of venous blood flow in the lower limb on duplex Doppler?
Respiratory rhythm generation center is located at:	Dorsal respiratory group Pre-Botzinger complex Ventral respiratory neurons Pneumotaxic center	Where is the respiratory rhythm generation center located?
A blue newborn presents with cyanosis. The X-ray chest reveals oligemic lung fields and a normal-sized heart. Most likely diagnosis is –	Ebstein’s anomaly Pulmonary atresia Transposition of great arteries Tetralogy of fallot	A newborn presents with cyanosis (appearing blue). Chest X-ray reveals oligemic lung fields and a normal-sized heart. What is the most likely diagnosis?
Polydactyly, craniosynostosis, Late closure of fontanelles is a feature of:	Apert’s syndrome Crouzon’s syndrome Pierre robin syndrome Down’s syndrome	Polydactyly, craniosynostosis, and late closure of fontanelles are features of which syndrome?
Low insulin to glucagon ratio is seen in all of these except:	Glycogen synthesis Glycogen breakdown Gluconeogenesis Ketogenesis	Which of the following is not associated with a low insulin-to-glucagon ratio?

Table 2: Comparison of Original and Cleaned Questions with Answer Options

2.3 Data for Experiments

We select a subset of the dataset that contains explanations. Then, we produce two different versions of this subset for training:

- Version **Original** consists of `exp` values, regardless of their formatting quality.
- Version **Cleaned** includes `exp` values in cases of good formatting and `exp_upd` values in cases when `exp` has formatting issues.

The original data contains around 20-21 million tokens, while the cleaned data is smaller, at 17-18 million tokens. These numbers are based on Figure 5 for the **Original** data and Figure 6 for the **Cleaned**. We estimate the number of tokens with model tokenizers. The corresponding models are introduced in Subsections 3.3 and 3.4.

INPUT		OUTPUT
Question	Original Answer Options	Cleaned Answer Options
What is the principle of Chinese medicine?	Yang Vin Both None	Yang Yin Both None
What visual field defect is typically produced by a temporal lobe tumor?	Crossed upper Quadrantanopia Uncrossed upper Quadrantanopia crossed lower Quadrantanopia Uncrossed lower Quadrantanopia	Crossed upper quadrantanopia Uncrossed upper quadrantanopia Crossed lower quadrantanopia Uncrossed lower quadrantanopia
Multiple canals in mandibular premolars are seen in?	Africas Caucians Not Recalled Not Recalled	Africans Caucasians Not recalled Not recalled
If the dentist concludes you don't have a cavity, when you really have one or more?	Type 2 error Type 1 error Type 3 error one of the above	Type 2 error Type 1 error Type 3 error one of the above
Which type of ZnPO ₄ cement is fine-grained, having film thickness of 25 micrometers?	Type 1 Type 2 Type 3 Type 4	Type 1 Type 2 Type 3 Type 4

Table 3: Comparison of Original vs Cleaned Answer Options

3 Methodology

This section introduces and motivates our domain adaptation approaches. Then, it gives more details on the model selection. For training, we use general-domain models that publicly provide their weights. We implement our solutions using the transformers library on Hugging Face² and PyTorch³.

3.1 Domain Adaptation

Domain Adaptation (DA) is a special form of Transfer Learning (TL) (Singhal et al., 2023). Basically, it can be seen as mimicking the human learning process. As humans, we tend to transfer our existing knowledge and experiences into solving new problems. Similarly, in DA, the goal is to take a model trained in one domain (the source) and achieve effective results in another (the target). Usually, the target domain is more specific and fine-grained. This is especially useful in practice, as high-quality labelled data is often scarce, so DA allows us to leverage the experience of a capable model and extend it to a new domain. Throughout the project, we use DA as a technique that enables our general models to

²<https://huggingface.co/>

³<https://pytorch.org/>

You are a careful and precise medical editor with deep experience in clinical texts. You refine questions from a medical multiple-choice dataset into clear, natural question format while preserving all essential context, details, and medical nuance.

You will be given:

- A possibly incomplete, fragmented, or awkwardly phrased question.
- Answer options (for reference only)

Your task: Return a polished, standalone question that

- Starts with a capital, ends with ?, uses proper grammar and medical terminology
- Includes all factual details, context, and scenarios from the original
- Adds appropriate question phrasing ("What", "Which", "Where", etc.) derived from content/options if missing
- Never removes or simplifies clinical context, patient descriptions, findings, or comparisons
- Copies well-formed questions verbatim unless minor polish helps clarity
- For "except"/"not"/"false" questions, preserve the negative logic clearly:
 - Original: "All are true except:" → Output: "Which of the following is ****not**** true?"
 - Original: "Test may be abnormal except:" → Output: "Which test is ****likely normal**** in [condition]?"
- Reference answer options **ONLY** IF they appear explicitly embedded in the question (e.g., "steps: a, b, c").
- Otherwise, **NEVER** mention, list, or reference options.
- Use options silently to derive question words only ("What", "Which", "Where").

Figure 1: Question cleaning system prompt

continuously learn from variations in the medical domain.

3.2 MLM vs CLM

Masked Language Modelling (MLM), as used in the original BERT paper (Devlin et al., 2019), is a powerful way to train any bidirectional model in language by performing a fill-in-the-blank task. Specifically, the MLM head corrupts the input training sequences with special [MASK] tokens, which are then predicted. Since this task is unsupervised, only raw text is needed. However, pretraining efficient models does not necessarily mean using bidirectional representations. Another successful language modelling objective is Causal Language Modelling (CLM). The models which employ this objective are called causal or autoregressive, as the generation of each next token is conditioned on what has already been generated. A popular example of a causal model is GPT-2 (Radford et al., 2019), which uses the exact same CLM. We use the CLM loss by simply asking the model to perform next-token prediction (NTP) at each step. Here is a simple illustrative example showing how MLM differs from CLM:

Consider the sentence: Is Attention All You Need?

Masked Language Modeling (MLM):

Input: Is [MASK] All You Need?

Target: Attention

Note: Model sees the full sequence

Causal Language Modeling (CLM):

Input: Is Attention All You

Target: Need?

Note: Model does not see the next word

You are a careful and precise medical editor with deep experience in clinical texts. You refine answers from a medical multiple-choice dataset into the correct format while preserving all essential context, details, and medical nuance.

You will be given:

- A question
- Possibly badly formatted answer options

Your task: return refined answer options that

- Start with option letter A., B., C., D., each answer option starting with a capital letter on the new line
- Keep the same order as the original, even if it's not logical
- Fix optical character recognition mistakes, use proper grammar and casing
- Use no punctuation at the end of the answer options, and ensure consistent formatting across all options
- Never change, add to, or remove from the original option text (including potentially incorrect medical terms)
- Include all factual details from the original options
- Copy well-formed answer options verbatim unless minor formatting polish helps clarity
- Preserve the original number of answer options and their order, even if they are repeated

Output only the refined answer options, each on a new line. Nothing else.

Figure 2: System prompt for cleaning answer options

We use both the MLM and CLM losses in our experiments and further select the specific models suitable for each group.

3.3 Bidirectional Models

Inspired by the newly learned non-standard attention mechanisms, we mainly focus on testing models on our data. However, we still include an empirically strong baseline that relies on full attention. Namely, our final bidirectional model list consists of:

- **RoBERTa** (facebookai/roberta-large) - RoBERTa (Liu et al., 2019) is a strong baseline which uses full attention and scales for sequences up to 512 tokens. We choose RoBERTa instead of BERT because of its more robust pretraining methods - more data and dynamic masking. Also, some of the models presented next are initialised from RoBERTa's checkpoints.
- **BigBird** (google/bigbird-roberta-large) - BigBird (Zaheer et al., 2021) belongs to the non-standard attention model group. It handles sequences of up to 4096 tokens, making it especially useful for tasks that require long contexts. It is based on Block Sparse attention, which, in contrast to full attention, does not scale quadratically with time. Moreover, by introducing a fixed number of global tokens (e.g., [CLS]), the quadratic dependency is reduced to linear.
- **LongFormer** (allenai/longformer-large-4096) - LongFormer (Beltagy et al., 2020) uses localised sliding window attention with a few task-specific global tokens and, as the name suggests, supports up to 4096 token inputs. It is shown to consistently outperform RoBERTa; however, not BigBird.
- **ModernBERT** (answerdotai/ModernBERT-large) - Originally, we aimed for RoFormer (Su et al., 2023), because of its contribution of the Rotary Positional Embeddings (RoPE). However, due to the inability to find an English-based checkpoint, we decided to continue with ModernBERT (Warner et al., 2024), which builds on RoFormer's main finding. ModernBERT is pretrained on sequences of length 8192 with alternating attention. It alternates between full attention, where

each token attends to all others, and local attention, where tokens attend only within a small sliding window. The local-attention mechanism is inspired by LongFormer.

3.4 Autoregressive Models

Similarly to the previous section, we include one standard full-attention baseline (GPT-2) and three variants. To employ the CLM objective for DA, we choose these autoregressive models:

- **GPT-2** (`openai-community/gpt2`) - GPT-2 (Radford et al., 2019) uses full attention with input sequences of 1024 consecutive tokens. Interestingly, GPT-2’s vocabulary is used by RoBERTa and, consequently, by BigBird. We choose this as our baseline model because of its size and the causal usage of the Transformer architecture (Vaswani et al., 2017).
- **Mamba** (`state-spaces/mamba-1.4b-hf`) - Mamba (Gu and Dao, 2024) is a structured state space model (SSM) that does not use the attention mechanism and scales linearly with input sequence length. Unlike Transformer-based models, Mamba models can be thought of as recurrent neural networks (RNNs) combined with convolutional neural networks (CNNs). Such models can use enormous sequence lengths of up to 1M tokens. We choose Mamba for experimentation because of its efficient yet non-standard architecture and its publicly available weights.
- **RWKV** (`RWKV/v5-Eagle-7B-HF`) -RWKV (Peng et al., 2023) is another RNN-based model. It employs Transformer-like training and RNN-like inference, both with linear time and space dependencies. We use the checkpoint with *flash-linear-attention*⁴. The **R**eceptance **W**eight **K**ey **V**alue model in theory does not have a predefined sequence limit, but in practice it is shown to be more useful when trained with continuously increasing sequence lengths (i.e. 1024 - 2048 - 4096 - 8192). However, we keep the sequence length static for DA.
- **KiMi Linear** (`moonshotai/Kimi-Linear-48B-A3B-Base`) - KiMi Linear (Team et al., 2025) is a much larger-scale model than the ones mentioned so far. However, we choose it because of its alternating Kimi Delta Attention (KDA) and Multi-Head Latent Attention (MLA). Kimi remains comparable to other MLA baselines on short sequences and shows visible advantages on longer texts. It is practically scalable for sequences of 1M tokens. Furthermore, it uses No Positional Embeddings (NoPE) for the MLA layers, while capturing positional information via its KDA.

4 Experiments

In this section, we introduce and explain in detail the designs for our bidirectional and autoregressive (causal) model experiments.

The experiments provide clear indications of how well the system adapts to the target domain. This is useful because it is a direct way to address our hypotheses.

To quantify domain adaptation, we track the model’s surprise when generating the data (i.e., perplexity) and its masked token prediction accuracy (masked accuracy) or next token prediction accuracy (NTP accuracy). For perplexity, the lower the better; for accuracy, the higher the better. Alongside these metrics, we also measure the runtime of the experiments.

Roughly, the experiments can be separated into the following groups:

- Fit on the `train` set - perplexity and accuracy on either original or cleaned data. The metrics are tracked at equal intervals and with a controlled number of samples, i.e., logging steps during training. However, we only report the final metrics on the full set.

⁴<https://pypi.org/project/flash-linear-attentiona>

- Evaluate on the `dev` set - perplexity and accuracy on the full dev set of the corresponding version. We initiate this procedure at the end of each epoch and again after training is over.

The intermediate results on the train split are reported in Table 5, and the evaluation results on the dev split are reported in Table 5.

4.1 Bidirectional Experiments

Table 4.1.1 shows the exact hyperparameters we used for the bidirectional models. Sequences are batch-dynamically padded with a 15% probability, which is standard in the literature. On the other hand, this slows down convergence. The maximum allowed length is deliberately set to 512, as this is the limit of RoBERTa-large. Also, this could be seen as a nice demonstration of whether non-standard attention models can be used as efficiently as standard full-attention models on shorter sequences. Our final train batch size is 32, and the evaluation batch size is 16. The learning rate allows for less aggressive adaptation.

Note that all bidirectional models are run on a single A100 GPU with 80 GB of RAM. The GPU had no other active processes at the time our experiments ran, ensuring fair measurement.

4.1.1 Downstream fine-tuning and evaluation

Additionally, we provide a small case study that removes the MLM head used for DA and only uses the encoder backbone. The case study focuses on fine-tuning a *MultipleChoice* model head to enable evaluation on the downstream NLP task, namely MCQA. The checkpoints of the adapted models are compared with their off-the-shelf counterparts. For our use case, off-the-shelf means directly using the publicly available model checkpoint from Hugging Face and fine-tuning on top. We compute macro averages for precision, recall and f1-score, and also include accuracy.

In Table 4.1.1, the hyperparameters for the case study fine-tunings are displayed. Again, they remain untouched across models for fair comparison. We decide to update the full model during fine-tuning, but only changing the learning rates for the backbone and the classifier head. The model is trained on the first 3000 annotated examples from the original version of the MedMCQA dataset and evaluated on the full original dev set questions.

4.2 Autoregressive Experiments

We conduct next-token-prediction experiments using the hyperparameters in Table 4.2. They are kept as close as possible to the MLM, except for the maximum sequence length (1024 instead of 512) and smaller batch sizes to compensate. The experiments are performed on the CLuster running on NVIDIA GeForce GTX 1080 Ti GPU with 48 GB of RAM. We cannot confidently state that this GPU was solely working on this task during our experiments.

The only finished experiment is with the GPT-2 model. Unfortunately, none of the other models ran with the provided resources.

Mamba could be run with the same environment as GPT-2. It could be fully loaded, but it would not run regardless of how small the maximum sequence length is, even as low as 64, yielding CUDA OutOfMemory errors.

With **RWKV**, we initially tried running `RWKV/RWKV7-Goose-World3-1.5B-HF`, but it seems so new that it is still unstable. The implementation provided at the `fla-hub/rwkv7-1.5B-world` checkpoint is furthermore not official. `RWKV/v6-Finch-1B6-HF` is incompatible with the recommended installation of the required `flash-linear-attention`, and `RWKV/v5-Eagle-7B-HF` only loads with its isolated installation. However, after loading, it results in the same CUDA OutOfMemory errors as Mamba. All attempted RWKV models needed a special environment. The authors

also recommend using their special `rwkv` library⁵, but we haven’t attempted it yet, as it would require rewriting the entire domain adaptation script.

KiMi Linear also needs a special environment. However, attempting to load it was unsuccessful, even with quantisation (which is not advised for training). The model would load at most to 50% and then cause a CUDA OutOfMemory error.

Therefore, our results in autoregressive models are limited and hinder our comparison.

5 Results & Discussion

Following the experimental design, which is described in Section 4, we now present and discuss the experimental results. To start off, Table 5 shows the outcomes of DA across all models after continuous evaluations on the dev set. The second column (Version) indicates the data version, either **Original** or **Cleaned**. For more details on the data, refer to Section 2.

We run all our bidirectional models for three epochs. However, we only show the results for the first and last epochs. In all cases, the last model checkpoint also has the best evaluation metrics. For us, the two most important dimensions are perplexity and masked accuracy. For **RoBERTa-large**, the last epoch on the cleaned dataset yields the lowest perplexity and the highest masked accuracy. **BigBird-RoBERTa-large** exhibits even larger differences between the fit on the original corpus and the cleaned corpus. Specifically, epoch 3 on **Original** has perplexity of 4.87, meanwhile the **Cleaned** goes down to 2.53. The accuracy is also the highest on the cleaned data. Overall, the cleaned data models consistently outperform the ones adapted to the original data. This observation aligns well with our hypothesis that models trained on the cleaned corpus will yield the lowest perplexity. It also speaks well to the valuable transformation we made to the original data. Generally, the trend shows a gradual, successful adaptation across all model instances. Based on the shown metrics, the cleaned version of **ModernBERT-large** has the lowest perplexity among all models after 3 epochs and also the highest masked accuracy. While it can handle much longer sequences thanks to its unique alternating (global/local) attention, **ModernBERT** demonstrates superior performance and a quick grasp of domain language, even under resource-constrained conditions.

On the other hand, the autoregressive model is run for more epochs (eight), so we report it in the results as well (see Table 5). **GPT-2** continues the trend of bidirectional models yielding the lowest perplexity in the last epoch on cleaned data. It is significantly higher than in any bidirectional model, and we also observe that training yields the highest gain (perplexity on the original data drops from 30.00 to 23.15), whereas in the other models, the change is moderate and closer to the optimum. However, we are deeply sceptical about the reported NTP accuracy for **GPT-2**. It is almost impossibly low, as if the model has no command of English, let alone the medical domain. Furthermore, it shows almost no improvement, even though the perplexity changes. We suppose that we might have incorrectly recorded it. Unfortunately, we had no chance to rerun the evaluation and confirm our idea.

Table 5 visualises the fits on the corresponding training datasets, as well as the running times of all models. The fastest one is also the one with the fewest parameters - **RoBERTa-large**. Again, **ModernBERT** seems to be the best learner, with a perplexity of only 2.23 on an about 18M-token corpus. Its runtime is also impressive, ranking it second-fastest after **RoBERTa-large** on the cleaned data. On the other hand, the slowest, **GPT-2**, is 7 times ahead of the next candidate, **Longformer-large**, even though it is 3.5 times smaller in terms of the number of parameters. Still, **GPT-2** also shows a drastic reduction in runtime for the cleaned data (by a factor of 6!), which we are reluctant to explain. We attribute **GPT-2**’s slowness to the classical attention mechanism and to the longer sequence length of 1024 used during training. Therefore, naïvely, if we normalise by size, **LongFormer** shows only moderate

⁵<https://pypi.org/project/rwkv/>

improvement compared to GPT-2. We noticed LongFormer purposefully pads its input IDs to a multiple of 512. We outline this as a limitation in the case study to be presented.

To investigate further, our small case study evaluates all *bidirectional* models on the downstream MedMCQA task. The results are shown in Table 5. There, `Longformer-large` is omitted precisely because it is inefficient with sequences of fewer than 512 tokens. As our question-answer pairs are generally short, LongFormer introduces additional computational overhead, which we find inefficient, so we decide to omit this model. Generally, the trend shows that our adapted models perform better than their off-the-shelf counterparts. Also, models adapted to cleaned data achieve better metrics than those adapted to the original data. The only exception is ModernBERT, but we should note that the results are visibly very close within this group.

Another interesting observation is that ModernBERT achieved nearly perfect scores on the training data after 4 epochs. For comparison, all other models barely minimised the loss. While on one hand we already have overfitting, on the other hand, one might argue for underfitting in the case of the other models. Eventually, choosing the right model for the dataset is also an important decision, especially when labelled resources are scarce.

6 Conclusion

After the experiments with the bidirectional models, we can revisit our hypotheses, now empirically supported. We found that **(1)** models with standard and non-standard attention achieve similar performance after domain adaptation. No significant differences in runtime have been observed, except for GPT-2, which is the slowest. However, LongFormer turned out to be quite impractical for us due to it enforcing longer sequences. This is something we understood through the case study. **(2)** The models adapted to the cleaned explanations yielded much lower perplexities and higher accuracies of predicting masked or next tokens than the ones on the cleaned data, and even trained faster. This speaks positively about the quality of our cleaned corpus. Furthermore, we demonstrated the effects of domain adaptation via fine-tuning on the QA task, and despite the limited training corpus size, we already observe that DA models trained on the cleaned data achieve the best downstream results in most cases.

7 Limitations

We were unable to run certain models due to their non-public status, the need for model-specific libraries, and resource constraints. Our series of autoregressive experiments was most affected.

We have not used parameter-efficient fine-tuning techniques like LoRA (Hu et al., 2021) for the larger LMs.

8 Acknowledgement of LLM Use

We declare that we used the following LLM-powered services when working on the project:

- Perplexity for searching for papers and LaTeX.
- Grammarly for editing: spelling and clarity.
- ChatGPT for rewriting Markdown tables to LaTeX.

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INPUT	OUTPUT
Original Explanation	Cleaned Explanation
Chronic urethral obstruction because of urinary calculi, prostatic hyperophy, tumors, normal pregnancy, tumors, uterine prolapse or functional disorders cause hydronephrosis which by definition is used to describe dilatation of renal pelvis and calculus associated with progressive atrophy of the kidney due to obstruction to the outflow of urine Refer Robbins 7yh/9,1012,9/e. P950	Chronic obstruction to urinary outflow, such as that caused by urinary calculi, prostatic hypertrophy, tumors, pregnancy, uterine prolapse, or functional outflow disorders, can lead to hydronephrosis. Hydronephrosis is defined as dilatation of the renal pelvis and calyces, associated with progressive renal parenchymal atrophy, resulting from persistent obstruction of urine flow.
Ans. (c) Vitamin B12 Ref: Harrison’s 19th ed. P 640 Vitamin B12 (Cobalamin) is synthesized solely by microorganisms. In humans, the only source for humans is food of animal origin, e.g., meat, fish, and dairy products. Vegetables, fruits, and other foods of nonanimal origin doesn’t contain Vitamin B12. Daily requirements of vitamin Bp is about 1–3 pg. Body stores are of the order of 2–3 mg, sufficient for 3–4 years if supplies are completely cut off.	Vitamin B12 (cobalamin) is synthesized solely by microorganisms. In humans, the only significant dietary sources are foods of animal origin such as meat, fish, and dairy products; fruits, vegetables, and other nonanimal foods do not naturally contain vitamin B12. The daily requirement is approximately 1–3 micrograms, while body stores are about 2–3 milligrams, which are sufficient for roughly 3–4 years even if dietary intake ceases completely.
Ans. A 45 yo male with DM presents with polyuria polydipsia weight loss. Random sugar 350 mg/dl. HbA1c 10.2%. Most likely diagnosis:	A 45-year-old male with diabetes mellitus presents with polyuria, polydipsia, and weight loss. Random blood sugar is 350 mg/dL, and HbA1c is 10.2%, consistent with uncontrolled hyperglycemia.
Hydronephrosis refers to dilation of the renal pelvis and calyces due to obstruction of urine flow, leading to parenchymal atrophy if prolonged. Robbins Basic Pathology, 9th ed., p. 950.	Hydronephrosis refers to dilation of the renal pelvis and calyces due to obstruction of urine flow, leading to parenchymal atrophy if prolonged. Robbins Basic Pathology, 9th ed., p. 950.

Table 4: Examples of Original and Cleaned Explanations. Problems are highlighted in orange.

Hyperparameter	Value
Train Examples	3000
Batch Size (Train / Eval)	8 / 8
Epochs	4
Optimizer	AdamW
Backbone Learning Rate	2×10^{-5}
Classifier Learning Rate	1×10^{-4}
Scheduler	Linear w/ Warmup
Warmup Ratio	0.1

Table 5: Hyperparameters for QA fine-tuning

You are a careful but fair medical editor. You review explanatory paragraphs from a medical multiple-choice dataset and decide whether they are acceptable as stand-alone educational contexts or whether they would need substantial editing or removal.

You will be given one explanatory paragraph.

Your task:

– Classify the paragraph as True if any of the following is present:

- It explicitly mentions the correct answer or answer option (for example: "Ans. A", "Answer: B", "Correct option: C", "Option D is correct", or the input starts from a stand-alone letter out of the option set).
- It contains serious formatting problems that would require extensive editing, such as:
 - pervasive grammatical or spelling errors that make the text hard to read or unprofessional,
 - heavily garbled or broken text (e.g., words run together, random line breaks, clear copy-paste artifacts),
 - very inconsistent capitalization of medical terms or sentence starts,
 - clearly excessive or misplaced spaces or punctuation throughout the paragraph.

– Classify the paragraph as False if:

- It does not mention the correct answer in any way, and
- It is generally well-formed and readable as a standalone explanation, even if it has a few minor issues (e.g., occasional typos, small punctuation mistakes, or slightly informal phrasing).

Output format:

Respond with exactly one word: True or False.

Do not provide explanations, comments, or any additional text.

EXAMPLES

Example 1:

(...)

Example 2:

- Input -

Ans: C. AzithromycinRef: n e u m o n i a - i n - a d u l t s - i n - t h e - o u t p a t i e n t - s e t t i n g H
4Only one drug which is active orally i.e. Azithromycin.We require OPD based treatment;
hence Azithromycin is the best answer here.

- Output -

True

(...)

Figure 3: System prompt for classifying explanations (shortened to save space).

You are a precise medical editor specialising in clinical explanations. You transform raw explanatory text from a medical QA dataset into clean, standalone educational contexts that fully explain the medical concept without referencing any question, answer choices, or "correctness."

You will be given one explanatory paragraph.

Rules:

- Remove ALL answer/question references ("Ans.", "correct", options)
- Fix grammar, formatting, flow
- Keep ALL original medical facts/numbers exactly
- Output ONLY the rewritten text

Figure 4: System prompt for cleaning explanations

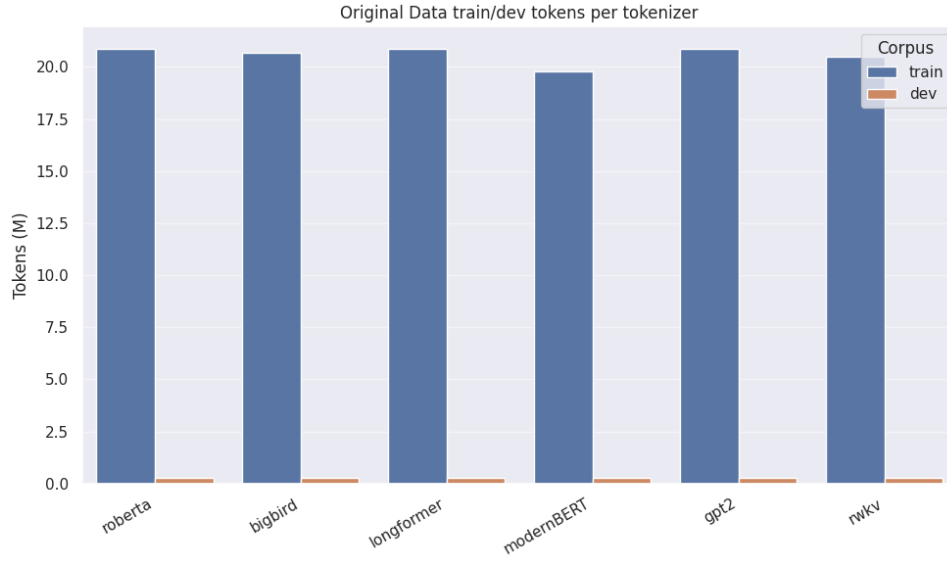


Figure 5: Original Data

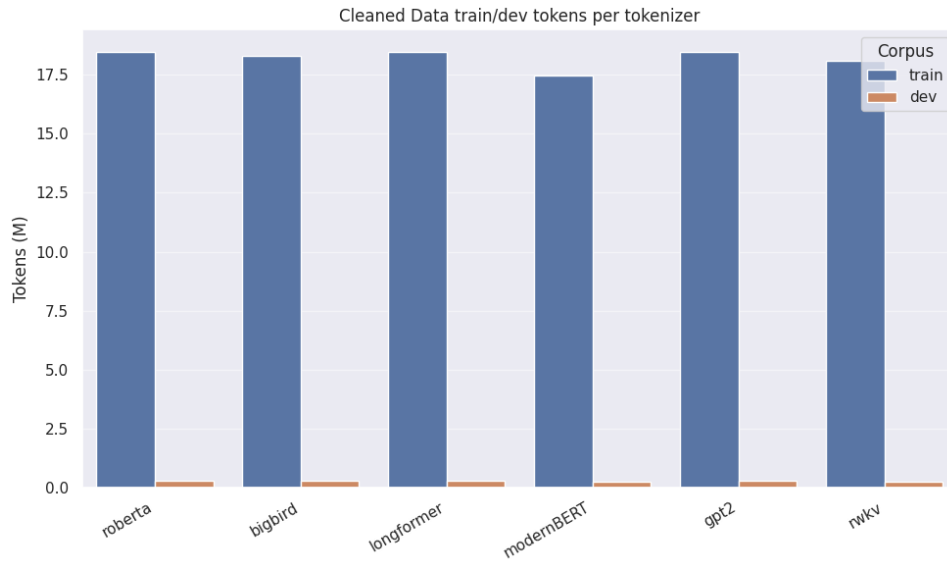


Figure 6: Cleaned Data

Hyperparameter	Value
MLM Probability	0.15
Epochs	3
Learning Rate	3×10^{-5}
Scheduler	Linear (warmup ratio = 0.05)
Weight Decay	0.01
Max Gradient Norm	1.0
Max Sequence Length	512
Train Batch Size (per device)	16
Eval Batch Size (per device)	8
Gradient Accumulation Steps	2
Dataloader Workers	4

Table 6: Hyperparameters used for MLM, for all models

Hyperparameter	Value
MLM Probability	0.0
Epochs	8
Learning Rate	3×10^{-5}
Scheduler	Linear (warmup ratio = 0.05)
Weight Decay	0.01
Max Gradient Norm	1.0
Max Sequence Length	1024
Train Batch Size (per device)	4
Eval Batch Size (per device)	1
Gradient Accumulation Steps	2
Dataloader Workers	4

Table 7: Hyperparameters used for NTP

Model	Split	Epoch	PPL ↓	Mask/NTP Acc.% ↑
RoBERTa-large	Original	1	4.23	71.58
	Original	3	3.74	73.35
	Cleaned	1	2.36	80.84
	Cleaned	3	2.20	82.36
BigBird-RoBERTa-large	Original	1	5.99	66.45
	Original	3	4.87	69.58
	Cleaned	1	2.81	78.06
	Cleaned	3	2.53	79.98
Longformer-large	Original	1	4.21	71.63
	Original	3	3.70	73.47
	Cleaned	1	2.35	80.67
	Cleaned	3	2.26	81.66
ModernBERT-large	Original	1	3.35	74.28
	Original	3	3.15	75.05
	Cleaned	1	2.03	83.27
	Cleaned	3	1.92	84.16
GPT-2 (base)	Original	1	30.00	1.84
	Original	3	25.17	1.80
	Original	8	23.15	1.77
	Cleaned	1	15.63	0.82
	Cleaned	3	14.31	0.80
	Cleaned	8	13.59	0.80

Table 8: Dev split DA results. Lower perplexity (PPL) and higher masked or next-token-prediction accuracy (Mask/NTP Acc.) indicate better performance. Bolded are the best metrics per model.

Model	Version	Epochs	Runtime (sec)	PPL ↓	Params (M)
RoBERTa-large	Original	3	8661	17.37	355.4
	Cleaned	3	7695	9.08	355.4
BigBird-RoBERTa-large	Original	3	10087	44.95	360.2
	Cleaned	3	9418	15.42	360.2
Longformer-large	Original	3	13350	16.89	434.7
	Cleaned	3	13081	8.54	434.7
ModernBERT-large	Original	3	9239	2.94	395.9
	Cleaned	3	8531	2.23	395.9
GPT-2 (base)	Original	8	94155	22.25	124
	Cleaned	8	15595	17.17	124

Table 9: Train split DA metadata and final train perplexity (PPL). Lower perplexity indicates a better fit on the training data.

Model	Backbone	P-macro	R-macro	F1-macro	Acc
RoBERTa-large	Off-the-shelf	23.23	23.12	22.94	23.17
	Original	23.49	23.56	23.49	23.88
	Cleaned	26.15	25.99	25.83	26.11
BigBird-large	Off-the-shelf	24.80	24.79	24.58	24.79
	Original	27.16	27.25	27.06	27.44
	Cleaned	28.35	28.28	28.10	28.42
ModernBERT-large	Off-the-shelf	28.32	28.44	28.01	28.11
	Original	27.52	27.72	27.39	27.54
	Cleaned	27.44	27.57	27.22	27.37

Table 10: QA performance on the original dev set of MedMCQA using 3k training samples. Longformer-large is excluded due to its inefficiency on short sequences.